Lecture 11. Markov Chain Monte Carlo

Markov chains, ergodic theorem, Metropolis–Hastings, Gibbs sampler

Contents

11.1 Markov chains **2**

11.2 The Metropolis–Hastings algorithm **3**

11.3 The Gibbs sampler **4**

11.4 Practical issues **5**

11.5 Selected exercises **6**

Markov chain Monte Carlo (MCMC) simulates a distribution π by running a Markov chain whose limiting distribution is π, and averages along the path. The draws are dependent, but an ergodic theorem replaces the law of large numbers.

11.1 Markov chains

A discrete-time Markov chain on a countable state space S is a sequence $\{X_n\}$ with

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \ldots, X_0) = P(X_{n+1} = j \mid X_n = i) = P_{ij}.$$

We assume time homogeneity (P_{ij} does not depend on n). The rows of the **transition matrix** $P = (P_{ij})$ sum to one. Examples: the simple random walk on $\mathbb{Z}$ ($P_{i,i+1} = p$, $P_{i,i-1} = 1 - p$); gambler's ruin on $\{0, \ldots, N\}$ with absorbing states 0 and N.

Chapman–Kolmogorov. The n-step probabilities $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfy $P^{(m+n)} = P^{(m)}P^{(n)}$ (condition on the state at time m), so $P^{(n)} = P^n$. If X_0 has distribution $\pi^{(0)}$ (a row vector), X_n has $\pi^{(0)}P^n$.

Classification.

- j is *reachable* from i ($i \to j$) if $P_{ij}^{(n)} > 0$ for some $n \geq 0$; i and j *communicate* if $i \to j$ and $j \to i$. Communication classes partition S; the chain is **irreducible** if there is only one. The random walk is irreducible; gambler's ruin has classes $\{0\}$, $\{1, \ldots, N-1\}$, $\{N\}$.
- The **period** of i is $d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}$; the chain is **aperiodic** if $d(i) = 1$ for all i.
- Let $\tau_{ii} = \min\{n \geq 1 : X_n = i\}$ given $X_0 = i$ (∞ if there is no return). State i is *recurrent* if $P(\tau_{ii} < \infty) = 1$, and *transient* otherwise; a recurrent state is **positive recurrent** if $\mathbb{E}\tau_{ii} < \infty$ and null recurrent otherwise. These are class properties.
- A chain is **ergodic** if it is aperiodic and positive recurrent. On a finite state space every state of an irreducible chain is positive recurrent, so irreducible plus aperiodic means ergodic.

Stationary distributions. π is **stationary** (invariant) if $\pi P = \pi$, $\pi \geq 0$, $\pi\mathbf{1} = 1$: started from π, the chain stays in π. If $P_{ij}^{(n)} \to \pi_j$ for all i, then π is stationary, since $\pi_k = \lim_n (P^{n+1})_{ik} = \sum_j \lim_n (P^n)_{ij} P_{jk} = (\pi P)_k$ (for finite S; in general by a dominated-convergence argument).

Theorem 11.1 (Limit theorems; stated without proof)

(a) An irreducible chain has a unique stationary distribution if and only if it is positive recurrent.

(b) If the chain is irreducible and ergodic, then $\lim_n P_{ij}^{(n)} = \pi_j$ exists independently of i, π is the unique stationary distribution, and $\pi_j = 1/\mathbb{E}\tau_{jj}$ is also the long-run fraction of time spent at j.

(c) If the chain is irreducible and aperiodic and a stationary distribution π exists, then $P_{ij}^{(n)} \to \pi_j$, and the chain is positive recurrent, hence ergodic.

Theorem 11.2 (Ergodic theorem)

If $\{X_n\}$ is irreducible and ergodic with stationary distribution π, and $\mathbb{E}_\pi|g(X)| < \infty$, then

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to \mathbb{E}_\pi g(X) \quad \text{almost surely, from any starting state.}$$

This extends the strong law from i.i.d. sequences to Markov chains and is the basis of MCMC: construct an irreducible, aperiodic, positive recurrent chain whose stationary distribution is the target f, run it, and average.

Detailed balance. The usual way to guarantee stationarity is the stronger **detailed balance** condition $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j.

Proposition 11.3

If π satisfies detailed balance, it is stationary. Moreover the stationary chain is *reversible*: $(X_0, \dots, X_n)$ and $(X_n, \dots, X_0)$ have the same distribution.

Proof. $(\pi P)_j = \sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \sum_i P_{ji} = \pi_j$. For reversibility, apply detailed balance repeatedly:

$$\pi_{i_0} P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} i_n} = P_{i_1 i_0} \pi_{i_1} P_{i_1 i_2} \cdots = \cdots = \pi_{i_n} P_{i_n i_{n-1}} \cdots P_{i_1 i_0}.$$

■

General state spaces. On a continuous state space, P_{ij} becomes a **transition kernel** $K(x, y)$, the conditional density of X_{n+1} at y given $X_n = x$, with $\int K(x, y)\, dy = 1$ (allowing point masses). Stationarity reads $\pi(y) = \int \pi(x) K(x, y)\, dx$; detailed balance $\pi(x)K(x, y) = \pi(y)K(y, x)$ implies it by integrating over x. Irreducibility, recurrence and the limit theorems carry over with suitable definitions.

11.2 The Metropolis–Hastings algorithm

Motivation. For complicated high-dimensional posteriors, independent sampling is impossible. Metropolis–Hastings constructs a Markov chain whose stationary distribution is the target, using only density ratios, so normalizing constants are never needed.

Let f be the target density, known possibly only up to a constant, and $g(\cdot \mid x)$ a **proposal** density that is easy to sample. Start at $X^{(0)}$ with $f(X^{(0)}) > 0$. For $t = 0, 1, \dots$:

(1) draw a candidate $\tilde{X} \sim g(\cdot \mid X^{(t)})$;

(2) compute the **Metropolis–Hastings ratio**

$$R(X^{(t)}, \tilde{X}) = \frac{f(\tilde{X})\, g(X^{(t)} \mid \tilde{X})}{f(X^{(t)})\, g(\tilde{X} \mid X^{(t)})};$$

(3) set $X^{(t+1)} = \tilde{X}$ with probability $\alpha(X^{(t)}, \tilde{X}) = \min\{R, 1\}$, and $X^{(t+1)} = X^{(t)}$ otherwise.

Metropolis et al. (1953) used a symmetric proposal, $g(y \mid x) = g(x \mid y)$ (e.g. a random walk $y = x + \epsilon$), for which $R = f(\tilde{X})/f(X^{(t)})$: uphill moves are always accepted, downhill moves sometimes. Hastings (1970) allowed asymmetric g. Only ratios of f appear, so its normalizing constant is never needed.

Theorem 11.4

The Metropolis–Hastings chain satisfies detailed balance with respect to f; hence f is a stationary distribution.

Proof. The transition kernel is

$$K(x,y) = \alpha(x,y)g(y \mid x) + \{1 - r(x)\}\delta_x(y), \qquad r(x) = \int \alpha(x,y)g(y \mid x)\,dy,$$

where $r(x)$ is the probability of accepting a move from x and δ_x is a point mass at x. For the continuous part,

$$f(x)\alpha(x,y)g(y \mid x) = \min\{f(x)g(y \mid x),\, f(y)g(x \mid y)\} = f(y)\alpha(y,x)g(x \mid y),$$

since the middle expression is symmetric in (x,y). For the point mass, $f(x)\{1 - r(x)\}\delta_x(y)$ is nonzero only at $y = x$, where it equals $f(y)\{1 - r(y)\}\delta_y(x)$. Adding, $f(x)K(x,y) = f(y)K(y,x)$, and stationarity follows as in Proposition 11.3. ■

Stationarity is automatic; the other conditions of Theorem 11.2 depend on g and must be checked:

- *Irreducibility* holds if $f > 0$ on S and the chain with kernel $g(y \mid x)$ alone is irreducible.
- *Aperiodicity* usually holds because rejections give $P(X^{(t+1)} = X^{(t)}) > 0$. For instance, in a discrete chain with $g(i \mid i) > 0$ and, from each i, some proposable neighbour j with $f(j) < f(i)$, we get $\alpha(i,j) < 1$, so $r(i) < \sum_{j'} g(j' \mid i) = 1$ and $K(i,i) \geq 1 - r(i) > 0$, whence $d(i) = 1$.

Example 11.1 (Posterior sampling)

For a posterior $p(\theta \mid y) = c\,p(y \mid \theta)p(\theta)$ with unknown c, the ratio is

$$R(\theta, \tilde{\theta}) = \frac{p(y \mid \tilde{\theta})p(\tilde{\theta})g(\theta \mid \tilde{\theta})}{p(y \mid \theta)p(\theta)g(\tilde{\theta} \mid \theta)},$$

free of c. With the **independence chain** $g(\cdot \mid \theta) = p(\cdot)$ (propose from the prior), the prior cancels and R is the likelihood ratio $p(y \mid \tilde{\theta})/p(y \mid \theta)$.
For example, with $X \sim \mathrm{Bin}(n,p)$, $p = e^\theta/(1+e^\theta)$, and prior $\theta \sim N(0,\sigma^2)$ (the model of Example 8.2), the posterior is proportional to $e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$, whose normalizer has no closed form. Proposing $\tilde{\theta} \sim N(0,\sigma^2)$ and accepting with probability $\min\{1, e^{(\tilde{\theta}-\theta)x}\{(1+e^\theta)/(1+e^{\tilde{\theta}})\}^n\}$ samples from it.

11.3 The Gibbs sampler

For $X = (X_1, \ldots, X_p)$ with a product state space, write X_{-i} for all coordinates but i. The Gibbs sampler updates coordinates in turn from their **full conditionals**, which are often standard distributions:

$$X_i^{(t+1)} \sim f_{X_i \mid X_{-i}}(\cdot \mid X_1^{(t+1)}, \ldots, X_{i-1}^{(t+1)}, X_{i+1}^{(t)}, \ldots, X_p^{(t)}), \qquad i = 1, \ldots, p,$$

always conditioning on the latest values (as in Gauss–Seidel, Lecture 5).

Example 11.2 (Bivariate normal)

For $(X_1, X_2) \sim N\big(0, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\big)$ the conditionals are $X_1 \mid X_2 = x_2 \sim N(\rho x_2, 1-\rho^2)$ and symmetrically. Hence $X_1^{(t+1)} \sim N(\rho X_2^{(t)}, 1-\rho^2)$, then $X_2^{(t+1)} \sim N(\rho X_1^{(t+1)}, 1-\rho^2)$. The marginal

chain $X_2^{(t)}$ is autoregressive with coefficient ρ^2, so mixing is slow when $|\rho|$ is near 1.

Proposition 11.5

Each coordinate update of the Gibbs sampler is a Metropolis–Hastings step whose proposal is always accepted. Hence f is stationary for the Gibbs sampler.

Proof. Fix coordinate i and the current point x. The update proposes $\tilde{x}$ with $\tilde{x}_{-i} = x_{-i}$ and $\tilde{x}_i \sim f_{X_i|X_{-i}}(\cdot \mid x_{-i})$, i.e. the proposal density is $g_i(\tilde{x} \mid x) = f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i})\,\delta(\tilde{x}_{-i} - x_{-i})$. On $\{\tilde{x}_{-i} = x_{-i}\}$, which has probability one, factor $f(x) = f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(x_i \mid x_{-i})$:

$$R(x, \tilde{x}) = \frac{f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i}) \cdot f_{X_i|X_{-i}}(x_i \mid x_{-i})}{f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(x_i \mid x_{-i}) \cdot f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i})} = 1.$$

By Theorem 11.4, each update leaves f invariant, and so does their composition. ■

So Gibbs is Metropolis–Hastings with a proposal that changes with the coordinate being updated (the composed chain is not reversible in general, but f is still stationary).

Example 11.3 (Ising model)

Place p particles around a circle (with $x_{p+1} = x_1$), each with spin $x_i \in \{-1, +1\}$, and let

$$p(x) = \frac{1}{Z} \exp\Big(\beta \sum_{j=1}^{p} x_j x_{j+1}\Big), \qquad \beta > 0,$$

which favours aligned neighbours. The normalizer Z is a sum of 2^p terms, infeasible for large p, but the full conditional of x_i depends only on its two neighbours and Z cancels. With $s_i = x_{i-1} + x_{i+1}$,

$$P(x_i = +1 \mid x_{-i}) = \frac{e^{\beta s_i}}{e^{\beta s_i} + e^{-\beta s_i}} = \frac{1}{1 + e^{-2\beta s_i}},$$

a logistic function of the neighbours. Equivalently, the current spin is flipped with probability $1/\{1 + \exp(2\beta x_i s_i)\}$.

11.4 Practical issues

- *Burn-in.* The chain starts away from stationarity; the initial portion of the run is discarded. How much is a judgement aided by trace plots and convergence diagnostics.
- *Mixing and stopping.* Strong autocorrelation (as in Example 11.2 with $|\rho| \approx 1$, or a random-walk proposal with a poorly chosen step size) means the chain explores slowly, and the effective sample size is much smaller than the number of iterations. The Monte Carlo standard error must account for the autocorrelation (e.g. by batch means).
- *Dependence.* To obtain nearly independent draws one can run N parallel chains from dispersed starting points and keep only the final values, or *thin* a single chain, keeping every k-th value $Y_m = X_{mk}$. Thinning reduces storage, not estimation error: for estimating $\mathbb{E}_\pi g$, the full chain is never worse.

11.5 Selected exercises

Motivation. A two-state chain shows the limit theorems in closed form; the bivariate normal Gibbs sampler quantifies slow mixing; and a short argument proves the claim above that thinning never improves an estimate.

Exercise 11.1 (The two-state chain)

Let $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ with $a, b \in (0,1]$, and put $\lambda = 1 - a - b$.
(a) Show that $\pi = \frac{1}{a+b}(b, a)$ satisfies detailed balance, hence is stationary.
(b) Show that $P^n = \Pi + \lambda^n E$ for $n \geq 1$, where $\Pi = \mathbf{1}\pi$ (both rows equal to π) and $E = \frac{1}{a+b}\left(\begin{smallmatrix} a & -a \\ -b & b \end{smallmatrix}\right)$.
(c) Deduce that $P_{ij}^{(n)} \to \pi_j$ geometrically fast unless $a = b = 1$, and explain the exception.

Solution. [Supplement]

(a) $\pi_1 P_{12} = \frac{ba}{a+b} = \pi_2 P_{21}$; apply Proposition 11.3.

(b)

Step 1. ($n = 1$.) Entrywise, $\Pi + \lambda E$ has $(1,1)$ entry $\frac{b+(1-a-b)a}{a+b} = \frac{(a+b)(1-a)}{a+b} = 1 - a$, $(1,2)$ entry $\frac{a-(1-a-b)a}{a+b} = a$, and similarly $(2,1)$ entry b and $(2,2)$ entry $1 - b$. So $P = \Pi + \lambda E$.

Step 2. (Products.) $\Pi^2 = \mathbf{1}(\pi\mathbf{1})\pi = \Pi$. The rows of E sum to 0, so $E\mathbf{1} = 0$ and $E\Pi = 0$. Also $\pi E = \frac{1}{(a+b)^2}(ba - ab,\ -ba + ab) = 0$, so $\Pi E = 0$. Finally $E^2 = \frac{1}{(a+b)^2}\left(\begin{smallmatrix} a^2+ab & -a^2-ab \\ -ab-b^2 & ab+b^2 \end{smallmatrix}\right) = E$.

Step 3. (Induction.) If $P^n = \Pi + \lambda^n E$, then $P^{n+1} = (\Pi + \lambda^n E)(\Pi + \lambda E) = \Pi + \lambda^{n+1} E$ by Step 2.

(c) By (b), $|P_{ij}^{(n)} - \pi_j| \leq |\lambda|^n \max\{a, b\}/(a+b)$. For $a, b \in (0,1]$ we have $\lambda \in [-1, 1)$, with $\lambda = -1$ only for $a = b = 1$; otherwise $|\lambda| < 1$ and the convergence is geometric with rate $|\lambda|$. If $a = b = 1$ the chain alternates deterministically between the two states: it has period 2, P^n alternates between I and P, and there is no limit, although $\pi = (\frac{1}{2}, \frac{1}{2})$ is still stationary. ■

Exercise 11.2 (Mixing of the bivariate normal Gibbs sampler)

In Example 11.2, with $|\rho| < 1$:
(a) Show that $X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t$ with $\varepsilon_t \sim N(0, 1 - \rho^4)$ independent of $X_2^{(0)}, \ldots, X_2^{(t)}$.
(b) Show that $N(0,1)$ is stationary for $(X_2^{(t)})$, and that started from it, $\mathrm{Corr}(X_2^{(t)}, X_2^{(t+k)}) = \rho^{2k}$.
(c) Show that for the stationary chain, $n \operatorname{Var}(\frac{1}{n}\sum_{t=1}^n X_2^{(t)}) \to \frac{1+\rho^2}{1-\rho^2}$, and interpret.

Solution. [Supplement]

(a) The two updates can be written $X_1^{(t+1)} = \rho X_2^{(t)} + \sqrt{1-\rho^2} Z_1$ and $X_2^{(t+1)} = \rho X_1^{(t+1)} + \sqrt{1-\rho^2} Z_2$, with $Z_1, Z_2 \sim N(0,1)$ independent of each other and of the past. Substituting,

$$X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t, \qquad \varepsilon_t = \rho\sqrt{1-\rho^2} Z_1 + \sqrt{1-\rho^2} Z_2,$$

and $\operatorname{Var} \varepsilon_t = (1-\rho^2)(\rho^2+1) = 1 - \rho^4$.

(b) If $X_2^{(t)} \sim N(0,1)$, then $X_2^{(t+1)}$ is normal with mean 0 and variance $\rho^4 + 1 - \rho^4 = 1$. Iterating (a), $X_2^{(t+k)} = \rho^{2k} X_2^{(t)} +$ (a combination of $\varepsilon_t, \ldots, \varepsilon_{t+k-1}$), and the noise terms are independent of $X_2^{(t)}$, so the covariance, hence the correlation, is ρ^{2k}.

(c) With $r = \rho^2 \in [0, 1)$ and unit variances,

$$n \operatorname{Var}\Big(\frac{1}{n}\sum_{t=1}^{n} X_2^{(t)}\Big) = \frac{1}{n}\sum_{s,t=1}^{n} r^{|s-t|} = 1 + 2\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big)r^k.$$

The terms increase to r^k as $n \to \infty$, so by monotone convergence the sum tends to $1 + 2r/(1-r) = (1+r)/(1-r)$. Thus n Gibbs draws are worth about $n(1-\rho^2)/(1+\rho^2)$ independent draws (the effective sample size); for $\rho = 0.99$ this is about $n/100$. ■

Exercise 11.3 (Thinning never helps)

Let (X_t) be a stationary sequence (the joint law of $(X_{t+s})_{t\ge 1}$ does not depend on s) and $\mathbb{E}g(X_1)^2 < \infty$. For integers $n, k \ge 1$ compare the full average and the thinned average

$$\bar{g}_{\text{full}} = \frac{1}{nk}\sum_{t=1}^{nk} g(X_t), \qquad \bar{g}_{\text{thin}} = \frac{1}{n}\sum_{m=1}^{n} g(X_{mk}).$$

Show that $\operatorname{Var}\bar{g}_{\text{full}} \le \operatorname{Var}\bar{g}_{\text{thin}}$.

Solution. [Supplement] For $j = 0, \dots, k-1$ let $A_j = \frac{1}{n}\sum_{m=1}^{n} g(X_{mk-j})$. The indices $mk - j$ ($1 \le m \le n$, $0 \le j < k$) run through $1, \dots, nk$ exactly once, so $\bar{g}_{\text{full}} = \frac{1}{k}\sum_j A_j$. A_j is the same function of the shifted sequence (X_{t-j}) as $A_0 = \bar{g}_{\text{thin}}$ is of (X_t), so by stationarity $\operatorname{Var} A_j = \operatorname{Var}\bar{g}_{\text{thin}} =: v$ for every j. By the Cauchy–Schwarz inequality $\operatorname{Cov}(A_i, A_j) \le \sqrt{\operatorname{Var} A_i \operatorname{Var} A_j} = v$, hence

$$\operatorname{Var}\bar{g}_{\text{full}} = \frac{1}{k^2}\sum_{i,j=0}^{k-1} \operatorname{Cov}(A_i, A_j) \le \frac{1}{k^2}\cdot k^2 v = v.$$

So keeping all nk draws is never worse than keeping every k-th one; thinning is justified only by storage or by the cost of evaluating g. ■